

The exact quantum maximum of the five-outcome CGLMP inequality

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Abstract

We give a computer-assisted exact certificate for the quantum maximum of the standard five-outcome Collins–Gisin–Linden–Massar–Popescu (CGLMP) inequality, in the normalization with local bound 2. The maximum is the largest real root μ of $5\mu^6 - 65\mu^4 + 144\mu^2 + 96\mu + 16 = 0$, namely $3.0157104755226732855\dots$. An explicit noncommutative sum of squares proves the upper bound for arbitrary local Hilbert spaces and arbitrary five-outcome local positive-operator-valued measurements. The proof first establishes a bounded commuting-projector operator inequality and then uses a fixed common embedding for both settings of each local measurement. A nonmaximally entangled state in $\mathbb{C}^5 \otimes \mathbb{C}^5$, with standard Fourier measurements, attains the bound. We give all exact coefficient data through five small Hermitian Gram blocks, distinguish quotient-ring identities from positivity in the selected embedding, and specify finite rational-arithmetic verification procedures. The result does not require convergence of a numerical optimizer or of a semidefinite hierarchy. It is a computer-assisted proof, not a proof-assistant formalization; no claim of uniqueness, self-testing, or absolute priority is made.

1 Introduction and relation to prior work

Bell inequalities compare correlations obtainable from local classical models with those obtainable from quantum measurements. The CGLMP family provides a two-party, two-setting inequality for each finite number of outcomes [1]. The number of outcomes and the dimension of a local quantum system are different parameters: establishing an optimum within a prescribed five-dimensional family does not, by itself, give a universal five-outcome quantum bound.

The standard Fourier measurement family and the improvement obtained with nonmaximally entangled states have a substantial history. Acín, Durt, Gisin, and Latorre [2] showed that nonmaximally entangled states give larger CGLMP violations than maximally entangled states with these measurements. Their Bell-operator calculation already reports 3.0157 for $d = 5$ in Table I; it does not supply the unrestricted quantum upper-bound certificate proved here. Zohren and Gill [3] studied the optimal-state structure and large-outcome behavior numerically. Hu, Deng, and Chen [4] investigated positive and negative violations by a Bell-operator method. Their numerical values and measurement candidates are background for the present work, rather than premises for its upper bound. In particular, the candidate decimal value reported here is not a newly discovered numerical prediction.

The Navascués–Pironio–Acín hierarchy provides semidefinite upper bounds on quantum correlations [5, 6]. Its dual perspective suggests noncommutative sum-of-squares (SOS) certificates. Once an exact positive certificate has been specified, its soundness can be checked directly, without rerunning the optimization that helped find it. Ioannou and Rosset [7] developed symmetry-adapted methods and gave exact CGLMP certificates for three and

four outcomes. Their version 2, Section VIII.4, reports a numerical five-outcome calculation and explicitly stops before an exact SOS decomposition; Section VII already describes exact candidate bounds and states. This comparison is specific to that version and those claims, and is not an assertion of exhaustive priority clearance.

The object supplied here is an exact five-outcome positive certificate with its actual algebraic embedding and a complete analytic connection to the local-POVM model. Its mathematical content is unchanged from the frozen scientific payload in [8]. This manuscript makes the coefficient data, root selection, positivity, physical attainment, and infinite-dimensional interfaces explicit in one place. The appendices contain the finite data needed to reconstruct the certificate without a numerical search.

2 Scenario, conventions, and theorem

Let $\mathcal{H}_A, \mathcal{H}_B$ be arbitrary complex Hilbert spaces, with no finite-dimensional or separability assumption. For $x, y \in \{0, 1\}$ let $\{E_{a|x}\}_{a=0}^4$ and $\{F_{b|y}\}_{b=0}^4$ be local POVMs: $E_{a|x}, F_{b|y} \succeq 0$ and each set sums to the identity on its own space. For a density operator ρ on $\mathcal{H}_A \otimes \mathcal{H}_B$, define

$$P(a, b|x, y) = \text{tr}[\rho(E_{a|x} \otimes F_{b|y})]. \quad (2.1)$$

The argument also holds for any normalized positive functional on the bounded-operator algebra, since the upper bound is an operator inequality. The symbols A_x, B_y below denote measurement outcomes, not Hermitian observables. All outcome equalities are modulo 5.

The functional is

$$\begin{aligned} I_5 = \sum_{k=0}^1 \left(1 - \frac{k}{2}\right) & [P(A_0 = B_0 + k) + P(B_0 = A_1 + k + 1) \\ & + P(A_1 = B_1 + k) + P(B_1 = A_0 + k) \\ & - P(A_0 = B_0 - k - 1) - P(B_0 = A_1 - k) \\ & - P(A_1 = B_1 - k - 1) - P(B_1 = A_0 - k - 1)]. \end{aligned} \quad (2.2)$$

For example, $P(A_x = B_y + k) = \sum_{b=0}^4 P(b + k, b|x, y)$.

Theorem 1. *Let μ be the largest real root of*

$$p(t) = 5t^6 - 65t^4 + 144t^2 + 96t + 16. \quad (2.3)$$

Every admissible strategy on arbitrary local Hilbert spaces, with an arbitrary normalized bipartite state and arbitrary five-outcome local POVMs, satisfies $I_5 \leq \mu$. Taking the supremum over all such local spaces, states, and POVMs gives

$$\sup I_5 = \mu = 3.015710475522673285523983801961\dots \quad (2.4)$$

The supremum is attained by the explicit projective strategy in Section 6 on $\mathbb{C}^5 \otimes \mathbb{C}^5$. The local hidden-variable bound in this normalization is 2.

For later use, put $[z]_5 \in \{0, 1, 2, 3, 4\}$, $f(z) = 1 - z/2$, and relabel

$$D_0 = A_0, \quad D_1 = B_0, \quad D_2 = A_1 + 1, \quad D_3 = B_1 + 1. \quad (2.5)$$

Directly comparing each pairwise event coefficient gives

$$I_5 = \mathbb{E} \left[\sum_{r=0}^2 f([D_r - D_{r+1}]_5) + f([D_3 - D_0 - 1]_5) \right]. \quad (2.6)$$

This notation is a sum of expectations from four experiments; it does not postulate a joint distribution for incompatible settings. For a deterministic local assignment, the four residues sum to a nonnegative integer congruent to 4 modulo 5, and hence to at least 4. Thus $I_5 \leq 4 - 4/2 = 2$. The assignment $A_0 = A_1 = B_0 = B_1 = 0$ attains 2, and convexity proves the local bound for arbitrary local randomization.

3 Bell polynomial and algebraic embedding

3.1 The universal projective algebra

Temporarily assume the measurements are projective. Let $\omega = e^{2\pi i/5}$ and encode the relabelled outcomes by unitaries

$$U_r = \sum_{a=0}^4 \omega^a P_a^{(r)}, \quad r = 0, 1, 2, 3. \quad (3.1)$$

The only relations used in the upper bound are

$$U_r^5 = \mathbf{1}, \quad U_r^\dagger = U_r^{-1}, \quad [U_r, U_s] = 0 \quad (r \text{ even}, s \text{ odd}). \quad (3.2)$$

There is no same-party commutation assumption. The discrete Fourier coefficients of f are

$$c_0 = 0, \quad c_k = \frac{1}{5} \sum_{z=0}^4 \left(1 - \frac{z}{2}\right) \omega^{-kz} = \frac{1}{2(1 - \omega^{-k})}, \quad 1 \leq k \leq 4. \quad (3.3)$$

Fourier synthesis in (2.6) yields the self-adjoint Bell polynomial

$$B = \sum_{r=0}^2 \sum_{k=1}^4 c_k U_r^k U_{r+1}^{-k} + \sum_{k=1}^4 c_k \omega^{-k} U_3^k U_0^{-k}. \quad (3.4)$$

In every projective realization its expectation is exactly (2.2). For exact checking, the finite sum in (3.3) avoids algebraic division and independently fixes the normalization and wraparound sign.

3.2 The selected actual root

We have $p(3) = -20$ and $p(31/10) > 0$, while

$$p'(t) = t^3(30t^2 - 260) + 288t + 96 > 0 \quad (t \geq 3). \quad (3.5)$$

Thus $(3, 31/10)$ contains exactly one root, and no larger real root exists. Set

$$s = \sqrt{5} > 0, \quad u = \sqrt{10 + 2s} > 0, \quad x = 5\mu. \quad (3.6)$$

The polynomial identity

$$[5(t^3 - 4t - 4)]^2 - 5(5t^2 - 4t - 8)^2 = 5p(t) \quad (3.7)$$

selects the positive square-root branch, since both bracketed quantities are positive for $t \geq 3$. Consequently

$$\mu^3 = s\mu^2 + (4 - 4s/5)\mu + 4 - 8s/5, \quad (3.8)$$

$$x^3 = 5sx^2 + (100 - 20s)x + 500 - 200s. \quad (3.9)$$

Define

$$\zeta = \frac{u + i(s-1)}{4}. \quad (3.10)$$

The relations in (3.6) give $|\zeta| = 1$, $\zeta^5 = i$, $\Im \zeta > 0$, and $\Re \zeta > 1/\sqrt{2}$. Among the fifth roots of i these conditions select $\zeta = e^{i\pi/10}$, so that $\omega = \zeta^4$.

3.3 Formal identities versus actual positivity

The serialized scalar convention is

$$\langle n; d \rangle = \frac{1}{d} \sum_{\substack{a,c,e \in \{0,1\} \\ b \in \{0,1,2\}}} n_{a+2b+6c+12e} s^a x^b u^c i^e, \quad d \in \mathbb{Z}_{>0}. \quad (3.11)$$

Every listed numerator is an integer. Products can be reduced with $s^2 = 5$, $u^2 = 10 + 2s$, $i^2 = -1$, and (3.9). This is a quotient-ring calculation. It need not assume that these relations give an irreducible field presentation: a zero obtained using them evaluates to zero at the specified actual complex numbers. Conversely, a formal zero does not select a real root or prove that a scalar is positive.

For a separate implementation one may use the 24 monomials $\zeta^a x^b$, $0 \leq a < 8$, $0 \leq b < 3$, and

$$\Phi_{20}(z) = z^8 - z^6 + z^4 - z^2 + 1, \quad s = 2(\zeta^2 + \zeta^{-2}) - 1, \quad u = 2(\zeta + \zeta^{-1}), \quad i = \zeta^5. \quad (3.12)$$

The same soundness statement applies. In particular, compatibility of a different embedding with the formal identity does not imply positivity in that embedding. Positivity below is proved only after the actual embedding (3.6)–(3.10) has been fixed.

4 An explicit finite sum-of-squares certificate

4.1 Generating words and exact Gram data

Index from zero the ordered lists

$$a = (\mathbf{1}, U_0, U_0^2, U_0^3, U_0^4, U_2, U_2^2, U_2^3, U_2^4), \quad (4.1)$$

$$b = (\mathbf{1}, U_1, U_1^2, U_1^3, U_1^4, U_3, U_3^2, U_3^3, U_3^4), \quad q_{9r+t} = a_r b_t. \quad (4.2)$$

Thus q has 81 entries with an unambiguous tensor-list order. There are five blocks, labelled $k = 1, 6, 7, 8, 9$, of sizes $n_k = 2, 4, 3, 3, 2$. Their labels are identifiers, not consecutive indices.

Appendix A specifies 81×4 phase matrices F_k , $4 \times n_k$ matrices K_k , and Hermitian $n_k \times n_k$ matrices H_k . Set $E_k = F_k K_k$ and

$$G = \sum_{k \in \{1,6,7,8,9\}} E_k H_k E_k^\dagger. \quad (4.3)$$

The exact coefficient identity is

$$\mu \mathbf{1} - B = \sum_{i,j=0}^{80} \left(G_{ij} q_i^\dagger q_j + \overline{G_{ij}} q_i q_j^\dagger \right). \quad (4.4)$$

All of F_k, K_k, H_k are printed in the appendix, including the integer numerators and denominators. Equation (4.4) therefore specifies a finite, reproducible algebraic claim, rather than an instruction to find a feasible matrix.

Lemma 2 (Exact certificate). *For the data of Appendix A, every H_k is positive definite in the actual embedding (3.6), and (4.4) holds modulo (3.2). Equivalently, there are exactly 14 positive weights and 14 explicitly determined polynomials such that*

$$\mu \mathbf{1} - B = \sum_k \sum_{r=0}^{n_k-1} d_{kr} (R_{kr}^\dagger R_{kr} + R_{kr} R_{kr}^\dagger), \quad d_{kr} > 0. \quad (4.5)$$

These are the same weights and polynomials as in the frozen compact certificate SOS14.json.

4.2 Exact identity check

Here is a terminating coefficient-wise verification of the algebraic part. Represent a word by an ordered list $(r_1, m_1), \dots, (r_\ell, m_\ell)$, meaning $U_{r_1}^{m_1} \dots U_{r_\ell}^{m_\ell}$. Move even-index factors to the left and odd-index factors to the right, preserving their relative order within each party. In each of the two lists, combine adjacent powers of the same generator modulo 5, deleting powers congruent to zero and repeating if a new adjacency is created. The result is a reduced word in $(C_5 * C_5) \times (C_5 * C_5)$. This normal form is unique. The involution reverses the list and negates powers before reduction.

Form G by exact scalar arithmetic and expand the right-hand side of (4.4). For each normal word w , collect the coefficient

$$\sum_{i,j: \text{NF}(q_i^\dagger q_j)=w} G_{ij} + \sum_{i,j: \text{NF}(q_i q_j^\dagger)=w} \overline{G_{ij}} - [w](\mu \mathbf{1} - B). \quad (4.6)$$

Reducing its scalar coefficients by (3.9) and (3.12) gives zero for all 1,681 distinct words generated by the full 81-word sequence. Words outside that universe occur on neither side. This computation never commutes U_0 with U_2 , or U_1 with U_3 . The supplied verifier uses the finite Fourier sum in (3.3) to construct its target independently of any stored target string.

For clarity, the algebraic statement can also be tested directly after forming (4.5): its 14 polynomials have 164 nonzero coefficients in total, and the union of expanded support with the Bell target has 273 normal words. The counts 273 and 1,681 refer to different representations of the same identity, not to incompatible universes of verification.

4.3 Strict positivity with rational enclosures

Let Δ_{kj} be the determinant of the leading $j \times j$ submatrix of H_k , with $\Delta_{k0} = 1$. Determinants involve only additions and multiplications, so their exact computation does not presume that any pivot is nonzero. Appendix B gives rational boxes for μ, s, u and strict rational enclosures for every Δ_{kj} . All lower endpoints are positive. Each determinant is checked to be exactly real by conjugation. Sylvester's criterion proves that each H_k is positive definite.

For completeness, interval arithmetic here has its elementary inclusion meaning: $[a, b] + [c, d] = [a+c, b+d]$, and interval multiplication takes the minimum and maximum of ac, ad, bc, bd . Scalar signs are handled by taking the smaller and larger scaled endpoints. Substitute the checked boxes into either the real tower monomials or the real and imaginary parts of the cyclotomic monomials, and sum the exact rational multiples. No decimal rounding enters the test. The boxes themselves are checked by $p(\mu_-) < 0 < p(\mu_+)$, $s_-^2 < 5 < s_+^2$, and $u_-^2 < 10 + 2s_- \leq 10 + 2s_+ < u_+^2$, with positive endpoints.

The positive leading minors now justify the unit-lower-triangular factorization $H_k = L_k D_k L_k^\dagger$. It is determined without choices by

$$d_{kr} = (H_k)_{rr} - \sum_{t < r} |(L_k)_{rt}|^2 d_{kt}, \quad (4.7)$$

$$(L_k)_{jr} = \frac{(H_k)_{jr} - \sum_{t < r} (L_k)_{jt} d_{kt} \overline{(L_k)_{rt}}}{d_{kr}} \quad (j > r), \quad (4.8)$$

with $(L_k)_{rr} = 1$ and $(L_k)_{jr} = 0$ for $j < r$. Indeed $d_{kr} = \Delta_{k,r+1}/\Delta_{kr} > 0$. Define

$$R_{kr} = \sum_{i=0}^{80} \overline{(E_k L_k)_{ir}} q_i. \quad (4.9)$$

Expanding (4.9) gives (4.5) from (4.4). The conjugation in this formula is important: it fixes the orientation of the Gram convention. Exact comparison with the frozen compact table

checks all $81 \cdot 14 = 1,134$ coefficient positions, including zeros, and all 14 weights. The smallest weight is the one labelled 9:1 and lies strictly between $3584887/10^{10}$ and $3584888/10^{10}$.

Remark 3. The five small blocks are positive definite. The 81×81 Gram matrix G , whose rank is at most 14, is only claimed to be positive semidefinite. The printed Gram data and the serialized SOS data are two linked presentations; agreement of one presentation with its target does not, by itself, authenticate every byte of the other.

5 From the certificate to arbitrary local POVMs

5.1 Bounded representations

Evaluate (4.5) in any bounded-operator representation of (3.2) on any Hilbert space. For every vector ξ ,

$$\langle \xi, (\mu \mathbf{1} - B)\xi \rangle = \sum_{k,r} d_{kr} \left(\|R_{kr}\xi\|^2 + \|R_{kr}^\dagger \xi\|^2 \right) \geq 0. \quad (5.1)$$

The sum is finite and its factors are bounded polynomials in unitaries. There is no truncation, domain problem, or interchange of limits. Thus $B \preceq \mu \mathbf{1}$ for arbitrary commuting projective representations, and in particular for all tensor-product projective strategies.

5.2 A common embedding for both settings

We give an explicit local dilation to avoid any setting-dependent state embedding. On one party's arbitrary Hilbert space $\mathcal{H}$, let $\{M_{a|x}\}_{a=0}^4$ be the POVM for setting $x = 0, 1$. Put $\mathcal{L} = \mathcal{H}^{\oplus 5}$, with coordinate projections Π_a , and define

$$V_x : \mathcal{H} \longrightarrow \mathcal{L}, \quad V_x h = (\sqrt{M_{0|x}}h, \dots, \sqrt{M_{4|x}}h). \quad (5.2)$$

Positivity gives bounded positive square roots and $V_x^\dagger V_x = \sum_a M_{a|x} = \mathbf{1}$. The defect $D_x = \mathbf{1}_{\mathcal{L}} - V_x V_x^\dagger$ is an orthogonal projection. On the *same* space $\tilde{\mathcal{H}} = \mathcal{H} \oplus \mathcal{L}$ for both settings set

$$Jh = (h, 0), \quad W_x = \begin{pmatrix} 0 & V_x^\dagger \\ V_x & D_x \end{pmatrix}, \quad Q_a = \begin{pmatrix} \delta_{a0} \mathbf{1}_{\mathcal{H}} & 0 \\ 0 & \Pi_a \end{pmatrix}. \quad (5.3)$$

The Q_a are five mutually orthogonal projections summing to the identity; their upper block assigns the unused summand to outcome 0. Moreover,

$$W_x^\dagger = W_x, \quad W_x^2 = \begin{pmatrix} V_x^\dagger V_x & V_x^\dagger D_x \\ D_x V_x & V_x V_x^\dagger + D_x^2 \end{pmatrix} = \mathbf{1}. \quad (5.4)$$

This uses $V_x^\dagger V_x = \mathbf{1}$, not the generally false assertion $V_x V_x^\dagger = \mathbf{1}$. It holds also for proper isometries in infinite dimension. The projectors $\tilde{M}_{a|x} = W_x Q_a W_x$ therefore satisfy

$$J^\dagger \tilde{M}_{a|x} J = V_x^\dagger \Pi_a V_x = M_{a|x} \quad \text{for every } a, x, \quad (5.5)$$

with precisely the same fixed J in both equations for x .

Apply this construction separately to Alice and Bob. With $\mathcal{J} = J_A \otimes J_B$ we have, for every a, b, x, y ,

$$\mathcal{J}^\dagger (\tilde{E}_{a|x} \otimes \tilde{F}_{b|y}) \mathcal{J} = E_{a|x} \otimes F_{b|y}. \quad (5.6)$$

Thus the single embedded state $\tilde{\rho} = \mathcal{J} \rho \mathcal{J}^\dagger$ preserves all joint probabilities simultaneously. Zero effects and mixed states cause no difficulty. For a general positive normalized functional φ , the formula $T \mapsto \varphi(\mathcal{J}^\dagger T \mathcal{J})$ gives the corresponding embedded state functional.

The local tensor-product construction preserves cross-party commutation, so the projective upper bound gives $I_5 \leq \mu$ for every POVM strategy in Theorem 1. Compression has only been used in (5.6), where the two local factors separate. We have not treated compression as a homomorphism on arbitrary products of same-party operators. Nor have we asserted a simultaneous dilation theorem for arbitrary abstract commuting POVMs: the POVM scope here is the stated local tensor-product model.

6 An explicit attaining strategy

Let $|j\rangle$, $0 \leq j \leq 4$, be the standard basis of $\mathbb{C}^5$. Use the projectors onto

$$|a; x\rangle = \frac{1}{\sqrt{5}} \sum_{j=0}^4 e^{2\pi i j(a+\alpha_x)/5} |j\rangle, \quad (\alpha_0, \alpha_1) = (0, 1/2), \quad (6.1)$$

$$|b; y\rangle = \frac{1}{\sqrt{5}} \sum_{j=0}^4 e^{2\pi i j(-b+\beta_y)/5} |j\rangle, \quad (\beta_0, \beta_1) = (1/4, -1/4). \quad (6.2)$$

For each setting these vectors form an orthonormal basis by the elementary sum of fifth roots of unity. Define

$$t = \frac{u}{2}, \quad f_1 = \frac{1}{t}, \quad f_2 = \frac{s-1}{2}, \quad f_3 = \frac{t}{s}, \quad f_4 = \frac{s+1}{2}, \quad (6.3)$$

and

$$a = \frac{\mu(f_1 + f_3) + 2f_1 f_2}{\mu(\mu - f_2) - 2f_1^2}, \quad b = \frac{2(f_2 + f_1 a)}{\mu}. \quad (6.4)$$

Since $\mu > 3$ and $0 < f_1, f_2 < 1$, the denominator defining a exceeds 4. All divisions are legitimate and $a, b > 0$. Numerically, $a = 0.7188787436302164244\dots$ and $b = 0.6605215707550941694\dots$. The state is

$$|\psi\rangle = \frac{|00\rangle + a|11\rangle + b|22\rangle + a|33\rangle + |44\rangle}{\sqrt{2 + 2a^2 + b^2}}. \quad (6.5)$$

We spell out attainment analytically. In an equality event, summing over the common outcome in the Fourier projectors kills matrix elements from $|ll\rangle$ to $|jm\rangle$ unless $j = m \pmod{5}$. Consequently the subspace $\text{span}\{|jj\rangle : 0 \leq j \leq 4\}$ is invariant under B . Substitution of (6.1)–(6.2) into (2.2) gives on this subspace

$$T_{jj} = 0, \quad T_{jl} = \frac{1}{2 \cos(\pi(j-l)/10)} = f_{|j-l|} \quad (j \neq l). \quad (6.6)$$

This follows by summing the four setting-pair phases; equivalently it is the five-point Fourier sum (3.3) evaluated with the stated shifts. The trigonometric values in (6.6) are exactly those in (6.3).

For $\gamma = (1, a, b, a, 1)^\top$, the independent row equations are

$$\mu = f_4 + (f_1 + f_3)a + f_2 b, \quad (6.7)$$

$$\mu a = f_1 + f_3 + f_2 a + f_1 b, \quad (6.8)$$

$$\mu b = 2f_2 + 2f_1 a. \quad (6.9)$$

Equations (6.8)–(6.9) give (6.4). The determinant of the reduced three-dimensional eigenvalue equation is

$$\det \begin{pmatrix} z - f_4 & -(f_1 + f_3) & -f_2 \\ -(f_1 + f_3) & z - f_2 & -f_1 \\ -2f_2 & -2f_1 & z \end{pmatrix} = z^3 - sz^2 - (4 - 4s/5)z - 4 + 8s/5. \quad (6.10)$$

At $z = \mu$ this vanishes by (3.8); the nonzero denominator already established implies (6.7). Hence $T\gamma = \mu\gamma$. Together with invariance this proves $B|\psi\rangle = \mu|\psi\rangle$ in the full 25-dimensional space. It proves $I_5 = \mu$ and completes the proof of Theorem 1.

The artifact additionally reconstructs all 100 projector-to-unitary matrix entries and checks the 25 full-space eigenvector coordinates exactly. These checks guard against phase, outcome-label, and tensor-order transcription errors; they do not replace the analytic universal upper-bound argument.

7 Computer-assisted verification and reproducibility

7.1 What is finite, and what is analytic

The load-bearing finite claims are the exact coefficient identity, Hermiticity and strict positivity in the selected embedding, and the exact attaining strategy. The bounded-representation implication and the common-embedding local-POVM bridge are analytic arguments given above. There is no inference from a finite sample of dimensions to arbitrary Hilbert spaces.

The discovery calculation used symmetry reduction, numerical exploration, and exact reconstruction. Five retained blocks produced 42 real parameters; the exact linear system had rank 34 and eight free parameters were fixed rationally. This history motivates the data but is not an assumption of the final proof. Appendix A supplies the resulting parameters directly; no search, solver status, numerical matrix-rank guess, or hierarchy-convergence argument is needed for verification.

7.2 Replay paths and their dependence

The frozen payload provides three relevant entry points:

- `verify_independent.py`: full 1,681-word Gram identity, exact LDL reconstruction, rational positivity, and physical attainment;
- `verify_sos14.py`: direct expansion of the 14 compact anticommutators, without reading Gram data;
- `verify_statement.py`: the root bridge, stated Schmidt amplitudes, denominator bound, reduced eigenvector, and Fourier orthonormality. The literal event convention is checked by the physical routine used in the first two entry points.

All three paths share scalar arithmetic in the distributed implementation; they are not separate independent votes. The name of an entry point does not establish independence of its author, runtime, or mathematical premises. The release-level hardened entry points add strict schema and target/input binding, without changing the frozen numerical coefficients.

The paper’s `check_paper_data.py` regenerates the printed data, checks the phase-factorization $E_k = F_k K_k$, reconstructs every Hermitian matrix, and verifies agreement with all canonical compact coefficients and weights. It computes the leading-minor intervals in Appendix B by determinant expansion and rational interval arithmetic. This is a data-correspondence check using the published exact arithmetic implementation, not a claim of a new independent implementation.

The repository preserves the frozen payload, historical receipts, and later adversarial-audit evidence separately. Subsequent multi-audit work tested literal events, alternate coefficient representations, same-party word order, actual branch signs, generic operator representations, and the fixed POVM embedding. Failed and partial audit runs are retained and distinguished from successful replays. Finite-field and floating tests are supplementary: they prove neither characteristic-zero identity nor real positivity. Agreement among audits is not itself an argument for the theorem.

7.3 Practical reproducibility

The four principal frozen certificate files are `EXACT_KERNELS.json`, `EXACT_SOS_CANDIDATE.json`, `POSITIVITY_CERTIFICATE.json`, and `SOS14.json`. Their release manifests identify exact bytes. Descriptive discovery status strings, cached pass flags, decimal spectra, and historical file pointers are not mathematical premises. A successful replay must check its intended semantic result, not merely a zero exit code. The release records sources, inputs, tool versions, and outputs, and rejects unsupported optimized-Python execution where assertions could otherwise be stripped.

The manuscript is buildable with standard pdfLaTeX and BibTeX; the source, complete coefficient appendix, bibliography, and clean build command are distributed with it. The full scientific artifact is available at <https://github.com/LeoLam233/cgimp5-exact-value> [8]. Hashes are provenance aids: they identify data and code but do not prove the truth of a mathematical assertion or certify an unobserved execution.

8 Limitations and authorship disclosure

This theorem concerns exactly five outcomes per setting. It does not establish a formula for all outcome numbers, uniqueness of the realizing measurements or state, a self-testing theorem, or an optimal statistical strength claim. The attaining state is nonmaximally entangled, but its existence does not classify every maximizer. The commuting-projector upper bound and the local tensor-product POVM theorem have been deliberately distinguished.

The proof depends on the correctness of explicit finite exact calculations and their elementary analytic interfaces. Public data and multiple reconstructions reduce specific risks; no finite testing campaign eliminates all possible software, runtime, or hardware errors. Neither this release nor the accompanying audit verdict is human peer review, formal proof-assistant verification, or an absolute originality/priority guarantee.

AI assistance and responsibility. Dehao Lin is the sole human author. Frontier AI systems substantially assisted with problem solving, proof construction, exact-certificate generation, verification software, adversarial auditing, literature triage, manuscript preparation, and release work. No AI system is a manuscript author. The human author selected and supervised the research workflow, decided what to preserve and publish, and takes responsibility for the public claims and artifact curation. Repository contributor attribution is distinct from academic authorship.

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A Complete exact coefficient data

This appendix specifies the certificate entirely by integers and the actual numbers of Section 3. All indices in the data are zero-based. The factorized representation is more compact than printing the expanded 164 coefficients of the LDL-derived SOS, some of which have several hundred digits. It changes neither those coefficients nor their weights.

A.1 Phase supports

Each row in Table 1 lists the four nonzero entries of column ℓ of F_k . A pair (j, e) means $(F_k)_{j\ell} = \zeta^e$; all unlisted entries are zero. The row index j refers exactly to (4.2).

Table 1: Nonzero phase-support pairs (j, e) in column ℓ of F_k .

k	ℓ	(j, e)			
1	0	(1, 0)	(5, 2)	(9, 19)	(45, 1)
1	1	(22, 0)	(42, 7)	(62, 2)	(74, 1)
1	2	(26, 0)	(38, 5)	(58, 6)	(78, 7)
1	3	(30, 0)	(34, 11)	(66, 1)	(70, 2)
6	0	(1, 0)	(5, 12)	(9, 14)	(45, 6)
6	1	(22, 0)	(42, 2)	(62, 12)	(74, 6)
6	2	(26, 0)	(38, 10)	(58, 16)	(78, 2)
6	3	(30, 0)	(34, 6)	(66, 6)	(70, 12)
7	0	(2, 0)	(6, 14)	(18, 13)	(54, 7)
7	1	(10, 0)	(14, 17)	(46, 7)	(50, 14)
7	2	(31, 0)	(43, 5)	(71, 14)	(75, 7)
7	3	(35, 0)	(39, 11)	(67, 18)	(79, 5)
8	0	(3, 0)	(7, 16)	(27, 12)	(63, 8)
8	1	(11, 0)	(23, 16)	(51, 16)	(55, 8)
8	2	(15, 0)	(19, 0)	(47, 8)	(59, 16)
8	3	(40, 0)	(44, 8)	(76, 8)	(80, 16)
9	0	(4, 0)	(8, 18)	(36, 11)	(72, 9)
9	1	(12, 0)	(32, 15)	(52, 18)	(64, 9)
9	2	(16, 0)	(28, 17)	(48, 6)	(68, 15)
9	3	(20, 0)	(24, 19)	(56, 9)	(60, 18)

A.2 Small kernel factors

Let $\langle n; d \rangle$ have the 24-coordinate meaning in (3.11). The matrices K_1, K_9 have their last two rows equal to I_2 , and their first two rows are supplied below. The matrices K_7, K_8 have their last three rows equal to I_3 , and their first rows are supplied below. Finally $K_6 = I_4$. Every coordinate is printed, including zeros. A line labelled $n[a:b]$ lists entries $n_a, \dots, n_{b-1}$; line breaks carry no further algebraic meaning.

$K_1[0, 0]$

```

d = 456608800
n[00:06] = (-183905100, -201247900, -137909750, -49175550, 7673370, 4177738)
n[06:12] = (68247450, 20591550, 19432545, 7021015, -1042545, -337979)
n[12:18] = (-101720500, -34720100, -45724350, -9485230, 248530, 387514)
n[18:24] = (86040650, -6912450, 16514365, 4426015, -980665, -259591)

```

$K_1[0, 1]$

```
d = 228304400
n[00:06] = (16933250, 28986350, -13620775, -1326085, 2374755, 832045)
n[06:12] = (4038550, -12864700, -14365490, -4193345, 333665, 128812)
n[12:18] = (-269277950, -33314850, -68067125, -29143135, 8258725, 3563555)
n[18:24] = (72093950, 2478800, 13797740, 3361605, -1793895, -718640)
```

$K_1[1, 0]$

```
d = 228304400
n[00:06] = (22756100, -92877700, -4433400, -170460, -699060, 413880)
n[06:12] = (12403950, -3975850, 1556920, -1574790, 153420, -10786)
n[12:18] = (204063500, 58514700, 61269800, 29408400, -4842420, -2206444)
n[18:24] = (30784100, 12406500, 16181145, 7285315, -1182135, -501157)
```

$K_1[1, 1]$

```
d = 456608800
n[00:06] = (-365083800, -7766400, -138101500, -60617840, 11548240, 4542004)
n[06:12] = (-17350650, -59665450, -34323185, -15510045, 2190215, 1095615)
n[12:18] = (-39403200, 150511800, 114618100, 37621400, -6459120, -4116452)
n[18:24] = (160719150, 1114950, 19091555, 12781095, -2436585, -420777)
```

$K_7[0, 0]$

```
d = 400
n[00:06] = (-2400, -800, -700, -320, 60, 24)
n[06:12] = (500, 300, 205, 85, -15, -7)
n[12:18] = (-600, -600, -300, -160, 20, 16)
n[18:24] = (200, 200, 95, 55, -5, -5)
```

$K_7[0, 1]$

```
d = 400
n[00:06] = (2900, 1300, 1000, 460, -80, -36)
n[06:12] = (-600, -300, -185, -85, 15, 7)
n[12:18] = (1500, 900, 500, 240, -40, -20)
n[18:24] = (-550, -250, -175, -95, 15, 7)
```

$K_7[0, 2]$

```
d = 100
n[00:06] = (900, 400, 325, 135, -25, -11)
n[06:12] = (-225, -75, -60, -25, 5, 2)
n[12:18] = (0, 0, 0, 0, 0, 0)
n[18:24] = (0, 0, 0, 0, 0, 0)
```

$K_8[0, 0]$

```
d = 200
n[00:06] = (250, -270, -90, -30, 4, 4)
n[06:12] = (0, 0, 0, 0, 0, 0)
n[12:18] = (0, 0, 0, 0, 0, 0)
n[18:24] = (-25, 25, 10, 0, 0, 0)
```

$K_8[0, 1]$

```
d = 200
n[00:06] = (400, -20, 60, 40, -6, -2)
n[06:12] = (0, 0, 0, 0, 0, 0)
n[12:18] = (0, 0, 0, 0, 0, 0)
n[18:24] = (-175, 65, -5, -5, 2, 0)
```

$K_8[0, 2]$

```
d = 1
n[00:06] = (0, 0, 0, 0, 0, 0)
n[06:12] = (0, 0, 0, 0, 0, 0)
n[12:18] = (0, 0, 0, 0, 0, 0)
n[18:24] = (0, 0, 0, 0, 0, 0)
```

$K_9[0, 0]$

```
d = 456608800
n[00:06] = (240067500, 102352700, 66559350, 36328590, -3873570, -1331474)
n[06:12] = (15825550, 73550250, 35881365, 15315975, -1997665, -1224703)
n[12:18] = (7763100, -30074900, -25358950, 2861330, -203190, 265402)
n[18:24] = (-16752550, 3233950, -7512955, 731575, -99705, -79263)
```

$K_9[0, 1]$

```
d = 456608800
n[00:06] = (7221100, -178266500, -131618450, -54794210, 7091910, 3325310)
n[06:12] = (21383350, -49300750, 5305245, -4555025, -287725, 121605)
n[12:18] = (-308864500, -90514500, -82301550, -36753110, 13455410, 6158450)
n[18:24] = (-100457950, -25109050, -30732945, -9485495, 3974165, 1679351)
```

$K_9[1, 0]$

```
d = 456608800
n[00:06] = (-344881200, -96629000, -75496000, -34664220, 5696740, 2221680)
n[06:12] = (25865150, 32173150, 24601565, 7974285, -1597195, -730311)
n[12:18] = (-208560000, -156097000, -88408000, -48452940, 8045180, 3217592)
n[18:24] = (-75377850, 50503750, 27199135, 10667495, -1151845, -1119881)
```

$K_9[1, 1]$

```
d = 456608800
n[00:06] = (-365083800, -7766400, -138101500, -60617840, 11548240, 4542004)
n[06:12] = (17350650, 59665450, 34323185, 15510045, -2190215, -1095615)
n[12:18] = (39403200, -150511800, -114618100, -37621400, 6459120, 4116452)
n[18:24] = (160719150, 1114950, 19091555, 12781095, -2436585, -420777)
```

A.3 Hermitian parameters

For each block, list the diagonal entries first, in increasing index, and then the upper-triangular pairs (a, b) in lexicographic order, giving their real and imaginary parts in that order. The 42 real parameters $h_0, \dots, h_{41}$ have the assignments shown in Table 2. An entry marked D means $(H_k)_{aa} = h_m$; R and I mean respectively $\Re(H_k)_{ab} = h_m$ and $\Im(H_k)_{ab} = h_m$ for $a < b$. The lower triangle is the conjugate of the upper triangle.

Table 2: Parameter assignments. Each half-row is an independent record.

m	k	a	b	type	m	k	a	b	type
0	1	0	0	D	21	7	1	1	D
1	1	1	1	D	22	7	2	2	D
2	1	0	1	R	23	7	0	1	R
3	1	0	1	I	24	7	0	1	I
4	6	0	0	D	25	7	0	2	R
5	6	1	1	D	26	7	0	2	I
6	6	2	2	D	27	7	1	2	R
7	6	3	3	D	28	7	1	2	I
8	6	0	1	R	29	8	0	0	D
9	6	0	1	I	30	8	1	1	D
10	6	0	2	R	31	8	2	2	D
11	6	0	2	I	32	8	0	1	R
12	6	0	3	R	33	8	0	1	I
13	6	0	3	I	34	8	0	2	R
14	6	1	2	R	35	8	0	2	I
15	6	1	2	I	36	8	1	2	R
16	6	1	3	R	37	8	1	2	I

m	k	a	b	type	m	k	a	b	type
17	6	1	3	I	38	9	0	0	D
18	6	2	3	R	39	9	1	1	D
19	6	2	3	I	40	9	0	1	R
20	7	0	0	D	41	9	0	1	I

Every parameter has the shorter real encoding

$$h_m = \frac{1}{d_m} \sum_{\substack{a,c \in \{0,1\} \\ b \in \{0,1,2\}}} n_{a+2b+6c}^{(m)} s^a x^b u^c. \quad (\text{A.1})$$

The following records print d_m and the twelve integers $n_0^{(m)}, \dots, n_{11}^{(m)}$ in order. For the purely rational entries the same convention applies, with all omitted coordinates explicitly zero.

h_0

```
d = 1366860518953188827148918565372465257414896071591421359075903127985169665347544101683200
n[00] = 245412364478276689360459046688661315178543575230279807345743091655159290164819596824500
n[01] = -11394429692409728839452319816213821915646969738792970951239751826723221876667160201180
n[02] = 5605816087487410185196761458480780438330946917235885747751828657912422364877761130930
n[03] = 31661967076866684482740854152755631664517588918033470626290613896232261564362786744530
n[04] = -5226858198576092061421936943422072296776596065093349856130426689931098926096022811224
n[05] = -1719723616535715924221694811500546307738765675892967468032880510409167111695756291136
n[06] = -38605834107952402683397329884440923240923908037272537967439572587716408396244269952150
n[07] = 45855208658212317706561133915065813009197002395469800959083781641986039384509353814170
n[08] = 26901175805644363896060261641575276826569802757407685639064153041889781405942706728415
n[09] = 9683295717139454151924490072394208679804053932207163567867179044891007633054807136435
n[10] = -1233350356995512951723070765905916006705664807375737928974630570635826964551872298230
n[11] = -1049091332021440928986200466351893255994130298396244054580933671270589306231841979944
```

h_1

```
d = 2147483648
n = (8225705, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
```

h_2

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = -38256020567417341341675935462905264097332864255380878686040986027665953187429911225200
n[01] = -24606636317723278746244325175715521490182605734409457281331215790702412580705344721360
n[02] = -60554357265422737177103669874149553352329903041547880640517248110054950862557931525080
n[03] = -28106984351329402664736409169512793971052066007698731331993030607179170521905989508280
n[04] = 4413427707546670504649328514582767041279239493047599573781384528512538456551163049412
n[05] = 1907539673572048163227221155676084906632126936325139347094329821457397427179335992312
n[06] = -44212618359121564358811850356015848982352863537135950066021767523854188050398232943100
n[07] = -12743989248689301359360486584387309602131602520587683467925602763179741358021880281460
n[08] = -31940318262999174876467224492355346155416458657343664985312116383261812196283222354550
n[09] = -13431330681908117164275203567081834201251756494773580281089428620499707082467663470770
n[10] = 2162100534425889937854093157603641648780116623498056984048582011195677377965316246685
n[11] = 1001955597509242099796060582059569919429599821161873314447451312427347954241591241525
```

h_3

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = 419844232216811423813115793811669844798897141823675698375824496757701896246588308124800
n[01] = 30602732617905862080206452147102474994091296844914153295507104501815697258355473849760
n[02] = 79831192706184075974801872078511085259433518473893702190482949571491310598596848890100
n[03] = 40611801169467291914847744761090599253997081241859863428698150308632666005519216736900
n[04] = -7455284672914820700599319752054110883475898464840922868610504796319836106166956731772
n[05] = -26743455288391116245145200992288746985064751927469096307125585390634156051123804328
n[06] = -24182000267372792064399782931127044108098835832959990080771322226080304048968841417700
n[07] = 45463681556841011658977730385142842743614474390280970761882119898378183899242043266580
n[08] = 4603572073360186226454672229629754366557429993684504456330604386161542631393146401960
n[09] = -193074976580091783317615567117509413882702734560802347658649687532051644390634221300
n[10] = -35658237387427591033925126504862628441273385735054396555433834115733564747658665709
n[11] = -265384743841121472141337163572908168796665491182932452130957252744629249741520905405
```

h_4

```
d = 4294967296
n = (501845005, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
```

h_5

```
d = 341715129738297206787229641343116314353724017897855339768975781996292416336886025420800
n[00] = -93212176226471784315346243448540559929781573142736056249018851934681535418905441421400
n[01] = -37440551519353710862408995506089595206083555776081770316545176523686031181754268862480
n[02] = -25344201712487537381589900793295065046574769329082571531232772907258850493444635493695
n[03] = -11992972421091229333263957115531600660200011667902366227216530794394681035167130734275
n[04] = 2090932940489596663134590636357937150161814468430108266393418332051713913225110915826
n[05] = 894032334677204332150915941475923370477422441172010071712996389426285014400070559512
n[06] = -20726383164457248371023186826754607060452955428765565093140707463428856793322457763700
n[07] = -10631034114272716935040580232479770182037436837895224175128912241958827012917469703410
n[08] = -4350803428849928180278364478869773820568238972306835999079431395417517862018113460285
n[09] = -1848570118441532880588348840435462602588315611712963648393759866728098538632274085035
n[10] = 438375764953295351443099316143887355415037405665708597169441270567905117620939555240
n[11] = 161725756314982541383946137710294653171640357058486212237190130971295558619718442369
```

h_6

```
d = 683430259476594413574459282686232628707448035795710679537951563992584832673772050841600
n[00] = -82813245166154708832530792263294858337154967914530818407238643214912365249392405920200
n[01] = -112727247288116951468484859652223835094220474241478008693022711737892627976435294631840
n[02] = -52163633280533279436566246868021086298825155525945100442418776995851283891938974425900
n[03] = -21123027888442893218012284526636632207824247009397506214731817839508550279748993786620
n[04] = 3389069931730817865653006893327215321063274459578432605717808341633093995057785072798
n[05] = 1955371905279809757547724663732558062461821980915374371023134086028305955355463478514
n[06] = -81380726078805152344178444828925233708139081557877321082959929039593559103860004898350
n[07] = -5637510202171919454390278658442929944542413341409305940996761993169988807750107641550
n[08] = -7769186134673623995441339772116722697915177268368749649817691352871339985133353177235
n[09] = -4693270051363230320989851053850162150334390163749034311331160295773416100135786128265
n[10] = 1169058040498867847511648937981339172690888030080882221980300999461714119039492277101
n[11] = 241947511099134816754551773183040713160661976060041431877147654019255366879786166501
```

h_7

```
d = 683430259476594413574459282686232628707448035795710679537951563992584832673772050841600
n[00] = 395280926859635089268287986122953562473577534344063262628042631127107059845859692244000
n[01] = 202045504877653826245704078244601253809701559157461368006861870684162863481805673870960
n[02] = 173185154483668746276181970756186079947428477596572793950067573631171625861338026066750
n[03] = 76948919657316977313367689998831652795894243016658504639882861733206992117144340039370
n[04] = -13605535734670015213895172119244421558827243279858456462998045889027072666316395979092
n[05] = -6067854269979903276078973862181132061117673023085767664554193459298329711890310045796
n[06] = 152003786003313447808793420385613513315762402568179271373568283548903999748792021314300
n[07] = 41080346192109271184710294659061440275348863412584052603094052612152746487579243195300
n[08] = 53742953141309790077928198076107108137353139255578571396508577600613548805649853610700
n[09] = 21857343706088014070490496918952873721913879266235851169180725805617860633164647504660
n[10] = -3919508062073923341517111566683315382619502949671728208536392467092888700431970178443
n[11] = -1703758227107723189941572893055163742852063087979235472354013635557531142341947506755
```

h_8

```
d = 4294967296
n = (-39622637, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
```

h_9

```
d = 2147483648
n = (-17650465, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
```

h_{10}

```
d = 2793135472143228293694245096652800
n[00] = 47835841622898479865456095142400
n[01] = 5493272592418579007118902408760
n[02] = 4066215439620936259187657286770
n[03] = 3831266685103962067343222653570
n[04] = -744356613373203440029482562062
n[05] = -323849719680936246223857542582
n[06] = -22478579129120105799812171276350
n[07] = 4615916590280372440756468702950
n[08] = -556272474815457729689662127855
n[09] = -1236347750335889564584875593545
n[10] = 327915155095382705206506573515
n[11] = 25985190790067278933108692577
```

h_{11}

```
d = 2147483648
n = (24295977, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
```

h_{12}

```
d = 2147483648
n = (-17690197, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
```

h_{13}

```
d = 4294967296
n = (-48696935, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
```

h_{14}

```
d = 1366860518953188827148918565372465257414896071591421359075903127985169665347544101683200
n[00] = 491821951719985230069443879781708176212733420554960115133841345419262842448534338427300
n[01] = 391966458315792933197496766401906700863722949365935764349969416363424225073274916892420
n[02] = 301256234711809971710813505121445251107358423477840672778260189363626312354935474133270
n[03] = 129173309607949308513722236976252976776571091748984855606724835628698343181097844670090
n[04] = -20971663187286138533175998314884294022055526213630269807080674073661533327391857753794
n[05] = -10386493618585598196666807950195989279621189514316448550886345627209297704027982731346
n[06] = 225888667140019892593042155692138293887962013307566903119903630530446436541845138370900
n[07] = 37908700072007305964663447672894843756603477579083858084612152763167042039559346395980
n[08] = 55550270252982232561800598649185918056200511357092145271711725045494868589691015560530
n[09] = 25070968389394611401262962913162787624591450748132234033179543400573364568071836239530
n[10] = -4898198755646738653034172716969288674509453750360464825232230159001885796701824059039
n[11] = -1840526206698916110324133784395320281915074966737362116697277295410924558339124573913
```

h_{15}

```
d = 1366860518953188827148918565372465257414896071591421359075903127985169665347544101683200
n[00] = -701169881350883339136082240743956064022576076654121133829045278183804642183475586615200
n[01] = -305981895736927934280400325350050764040076937791788385416993795961842352203302128545040
n[02] = -322373980083975407649029745562292559020006643262354285101265367243473986946607628524940
n[03] = -145416338438313811820022023366501584844735710021704450895359533131805745844932046532460
n[04] = 23834704128278657859159462713747361172142627303509887718205894277112014118672437626838
n[05] = 10943123393189751123205138157425143401219284167683889225012605796971175740850000541582
n[06] = -176698055856078159586379366804295204309672878988334705582068303766714329462069869416250
n[07] = -89845631121011001067129858260577779179184449122791608454148921718139758366437747165350
n[08] = -88979504489797271035899026375392859741256986121016026690198645668090324620392038975460
n[09] = -37749366648935951105950978381035589335958050493143073275220756288556885716336916329050
n[10] = 6462988556525557135082145361250342666575314533312194968350987501073163694599824540649
n[11] = 3080182308820291525720808492637703706105361340643801865062432764188636510535112892539
```

h_{16}

```
d = 1366860518953188827148918565372465257414896071591421359075903127985169665347544101683200
n[00] = -293648548014998808178401895229195188994139374251549960306003719060731006360240909121850
n[01] = -64440782780400855492497328077648243319858115870263257675019157911420090162664870713990
n[02] = -76400626051462657133363682218541365934785412993732449097691507375093979788022878264695
n[03] = -34997030896547601817560935007135811351024045639518675432216230858476424686614518325705
n[04] = 6833876329074046562490838720378342154030942108031880924098559086509730518492789668433
n[05] = 2776286273956831781476907972703293691098915074610924755143590359469417260184958456235
n[06] = -34719511464791184178908990098975510729971531290033933150606959143331581482894774804650
n[07] = -23718425791162730557130778253663049973389213523776108056725968044781445924469524389360
n[08] = -8824663749148716559693550171113744627761146661750150678403587285232987721685757244505
n[09] = -2221391246295701535939451368526492102010124543742593688520109904225173894158167302960
n[10] = 436418029367253624903500102374836330250440155972559795625713506180864923291584007586
n[11] = 286535796221470885157363030910618814751463590437079463637759452609837078163652559543
```

h_{17}

```
d = 1366860518953188827148918565372465257414896071591421359075903127985169665347544101683200
n[00] = 704487079928962447252273807350476610302335617985672392059171714294392208670796192403650
n[01] = 215280937775993329145406025873464891268060881126252658291206624383265586771850575132830
n[02] = 25804953346397320050810434000805371114653720193372322850712078595150146705730201012305
n[03] = 121737155934141381553582116030735994968204128690047416801834169304041104917155808943975
n[04] = -21152292729336375714064715190591200371366900263312796120377409303603674197739672585131
n[05] = -8717995187861924567890250886078635944796785881921510109055011462209230194554950462545
n[06] = 93233095259324286549192837682886936467158221385026085967912511099918618188626849990750
n[07] = 95018019867110961974560408422078036226296011194495341823299450379975383897626725598760
n[08] = 70677462892574006265995604971217947745922753395038946690322511180410096136229280275135
n[09] = 31114694028205426076220744387996342515732501853468311032453744891743306601841845041810
n[10] = -4997114845233648960711746350821699917036222782105640770097246236046739685283823205752
n[11] = -2492668135280212470632317338653195315259108075197621056651502284415109055378495291713
```

h_{18}

```
d = 1366860518953188827148918565372465257414896071591421359075903127985169665347544101683200
n[00] = 289837895602342935367166540911987810456706027552505472752243882164674783511641391554450
n[01] = 66311959030153299885848170978575168388516926613721588300611860093195157542004365684510
n[02] = 78261923961813475654599866165394096475508904068442078745277496495017818547300628678135
n[03] = 34940874102476541928788225558631821071024793095602389107651857549826137714130148820365
n[04] = -6796859062051714835640144604915000233957349709188068250066867998915772388625779560697
n[05] = -2848835304525121936466544613904776465765568277434587890679837943942673666224659959319
n[06] = 38673379646690193170074167540314996712760192381710603836575273353708121745635186371300
n[07] = 21906303454447974971452232233129905338259244477607381372893421180862339615809036237630
n[08] = 8017814461449459370302171746063288561567462454113207297923947437342603742594121299765
n[09] = 2378178214838578429203396434447919509874818297275400009343415142700955470525688198290
n[10] = -471160078230348891121105283533177014129862517045967536909147693659556052889036022050
n[11] = -256279679297541068734814130344307091878015975138899726326201498290608482512629515657
```

h_{19}

```
d = 1366860518953188827148918565372465257414896071591421359075903127985169665347544101683200
n[00] = 507153726821380688223057687137540022360364471407773717535690271416863228577195683920150
n[01] = 283046041768009977951679207002782068848650946086926768100153096064910857443325137836130
n[02] = 258092712763633830624398433235594412777077187031456127790728201262373613703001735797655
n[03] = 115470538676917693114094925155392270065188777606512408876293727523922131688687710491545
n[04] = -19537484576626436751502415032259450588995179087244827008678620231234355210108735266701
n[05] = -8926214332351986306240355680816690531745488846137868996336763127986115780180618187191
n[06] = 174794051483108262589507981551482777704737012125033756107497935090215255438122440970450
n[07] = 63927666587664211760086815606017447080809097385734175748745241289107963405230015048140
n[08] = 69761869221950369037138543668176297365308109756794205265233275530114907397453894492300
n[09] = 33166361756681887316258396828933015756740127882392914669805684579351166381626075082295
n[10] = -5621381934454402294718865621121137306422317204026373785618083807290002682191642329133
n[11] = -2348585847556249265506878394444337405721048507247443281453563730116445724673684702950
```

h_{20}

```
d = 1366860518953188827148918565372465257414896071591421359075903127985169665347544101683200
n[00] = -235982689539709975972991465445007491550745230253188468677834392359208605474052005642700
n[01] = -164552547714143411506728270696638716152862363769146482162759461935357887521215447929100
n[02] = -115026484076358557141398516482095289974940960111404727971975811276212185765203277500870
n[03] = -45406665759829249727064809883614224725355289654813822575992874948783748147570751747870
n[04] = 8403057552118811608509887009003120138005627004253164762839887460970105502899456977770
n[05] = 4455395253145158237681350227826603320253844388103827322631047603106423215184838858738
n[06] = -112555937775662311162696769785031672069306921498997614642240548750755974013411441501550
n[07] = -12345442507332585503562864811002971548844180255758377921632756344033911115118078036490
n[08] = -54962554299837920387718071638899590782703880258150495883721863869035353137891049226805
n[09] = -22502308497553752662064734161914421450988809732737277833368503669750173127993161453495
n[10] = 3577349163265773503047826141361266901375947402153818478014443139260590576132012224691
n[11] = 1640917197794259429716225825129027098875393705512955962813041207391469670287038035345
```

h_{21}

```
d = 8379406416429684881082735289958400
n[00] = 296050830919368131257441869618650
n[01] = -151732213946955710742508975434050
n[02] = -37552369609800877309515350858775
n[03] = -19239973838604527732732026729965
n[04] = 1771162365165672556769707065405
n[05] = 1802110731117904800417567881591
n[06] = -107151074367379930925675642768850
n[07] = 57518642834688666046406695661800
n[08] = 10344738626542428095901441853590
n[09] = 4845764606519362846077015943335
n[10] = -460005211828225354299233114600
n[11] = -476204889207908603032733650067
```

h_{22}

```
d = 4294967296
n = (19694267, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
```

h_{23}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = 243724321419008311235385953077050887168464483820632996153070523023829526776764263685100
n[01] = 199146037199101695294576522333348032267329654498498501534842764254632528988952757336300
n[02] = 99552237782962777945734446575668234192316147426865165066142025736453823983509995947400
n[03] = 41834960618374495916923448606135598267214692703656650871110670698147980108871831462700
n[04] = -7521987311761556960596494538087691612338364731092313670582138591537998935526070960760
n[05] = -3738239736074585498384047768421480280712283333802187493597341120949346362748840653136
n[06] = 171819277401554044884462553306513336223771118192081862103303679190458550351539572469550
n[07] = 28382680992492032227575868910420724874864895485832845639664530387481429244630727082770
n[08] = 22718160367973672521699543570581690080529234476881013406836945389499578569514041890505
n[09] = 11408442049685906353588991812702818440997204486528343055042288115232218634879338057105
n[10] = -2472727747014319016917806964329164875742922562499039620066170921204923787024848410216
n[11] = -783120868471591524536529034452981675286819806984698515943462603720103226019315517636
```

h_{24}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = -65918862728339347051273359249964190689316429477173543070569603661454682723144701270300
n[01] = -259816354186404115859253078296752440755848093868815214321588302892663690875074538393260
n[02] = -191920939487370320371489209019022172362102112967996585430210831768122114773561538100820
n[03] = -91802805894742434779527963429813807393066687458554366569460166694873153403719189608840
n[04] = 15853136588607838984393618921762867076111076420295619602206316508024707173268052486152
n[05] = 6623424020982660277730332140792015589887488810705752924452205083895036950582903034424
n[06] = -145950948389173562104132836109465659836888028605918061856359150113004781191759697996150
n[07] = -81151669743461939948455863961458246302868479179168668437820775234253671339549240852210
n[08] = -59630391970241955172230381490731748445658877169359429225009077650236727470371426246465
n[09] = -2518077749699279537523106581877820357882199977503636714867896861643907090539943585405
n[10] = 4549894098731050180205023458057994500967369588491081787207102240086730571663703467830
n[11] = 2179125837240238013749139260713042491567159230439507073333272015833025166955934105010
```

h_{25}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = -250553033582924209256426150408920727144255746910931137716500667781517906788371960996300
n[01] = -97055588281364852536194070669251333826434367052840461602312372905146990344534161130540
n[02] = -17337300432961584193387031954335202013680017342179711669289307701612185497251890210030
n[03] = -10505683296377681936667029911080871395620173770369658306833325674104549980695497813730
n[04] = 2844762035474905405786361689229583142869223892718926593102662634174708324489136589128
n[05] = 1063172078339764659609138723709120920926405368353453786134236967761129962828389432236
n[06] = -29904819888617837280643272760277470710587632606086009957824265433726143546325984113550
n[07] = -1846920385583991235781787708722163021080411491938790863397166656521711376209480508690
n[08] = -7757097367572246748080024992388030286025761089684671175223220740516952779436030239230
n[09] = -3381842134154483737490474373055556391581940447114444565278417382369610362436524478160
n[10] = 577996338716248034560052937993898805727633763740820235180786510220097655764940592215
n[11] = 283582895246441484159006166620564107181528025983396965581205966817865967248801328455
```

h_{26}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = -562830155022568897629358425563413156252792496788415731847626867032696421595723257980100
n[01] = -359032342063169199247281779889368672817166625430337352237955892000102703424025166342100
n[02] = -236543800968016981470324478702398801816351639885427629898783024877299966028111582932950
n[03] = -102564360167434091233976295383880572925689476951728479201729976983696404315125620328090
n[04] = 18191253781606707323580234753220770489511999717885174839086578303344937588652328117020
n[05] = 8808866076212798450921021798204154780332499483039180356351205710651551107607143721576
n[06] = -2505748909392578646257638023349127854689190394191072412895776910638528954984712229850
n[07] = -62853693558685941733355135881914871359766790054827194086270554223273514049683018252990
n[08] = -56300583958326366899116907744908144081780983129717066022495356213888842138904176950110
n[09] = -27452034649086269009397842661534618903825940686188064704134986125021360718065460810300
n[10] = 4927128202460291736655595430187529563121710910130799656874272994431631251612337015725
n[11] = 1939040937199528593396101691355621151103612660494924596578833331471793275933020478657
```

h_{27}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = -2169085024488417649901865659703565209907645304173251361583994403858819489934943850717100
n[01] = -1021473009998123196012974458889945618030638825313803213358792758599000732841411000982300
n[02] = -748165241270811700358680859795605328036721354910894823732917904454286687508744953254750
n[03] = -346987449762530292497210588365813313531968224721285089602529211575120694714459218687270
n[04] = 60945677283331825058186750224962731484440651461644897793064068908920851909605351657290
n[05] = 26753113812511773997855959862625635363734638561330330179303522774044011308110251592042
n[06] = -565268468251931844765221740909132618519472523320346385351449945766214220021749067832650
n[07] = -270880509254750400217510676139370956690757901860328414147873665186986197568134415187750
n[08] = -207182615826336156304479034060747723787412768434695140340268965947679130655925597994865
n[09] = -97808533790309628480408859627749453505348095431899186439604349599131887212594072074295
n[10] = 16985157364679875615315982744048857932392096574002929460657775014307202363761195376649
n[11] = 7329612373738830774395503298504145590971351169958801550526933162907716820352127681771
```

h_{28}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = 703674474812469337837034920473801212020224723393024733518630695257242645689760171878900
n[01] = 377431124057514078684655770621443185490614934522567563087047469879112364383582125677300
n[02] = 283307000000753038114030280079713994997088276072112100306339922956060879049173632010530
n[03] = 124576817962959132327063908833614890027299592290158032118791728332032038593751105989090
n[04] = -21501592834756657098796414920168205297962140897051539133476201493268928826721979666630
n[05] = -9994372573992388509312464552850544255506299933350815391982605176907736103618906434606
n[06] = 212342936250949383107560442468479860534292216435332913564712781331782940842005398686350
n[07] = 72780105256824095732142185833875073186281030608148404495630471835959458090443305486250
n[08] = 74989220926855767446876191295393862146408750233179873553154005292538967670391963756855
n[09] = 34190324350208904394272436161120939015139882025364810625564396112599032935263700791845
n[10] = -6042518418324536064268547612114308782062374041819670228394847975747865604314060755127
n[11] = -2551673843885838760627063095913947032159703430800600925808470636724398511839040153297
```

h_{29}

```
d = 4294967296
n = (19694267, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
```

h_{30}

```
d = 8379406416429684881082735289958400
n[00] = 296050830919368131257441869618650
n[01] = -151732213946955710742508975434050
n[02] = -37552369609800877309515350858775
n[03] = -19239973838604527732732026729965
n[04] = 1771162365165672556769707065405
n[05] = 1802110731117904800417567881591
n[06] = -107151074367379930925675642768850
n[07] = 57518642834688666046406695661800
n[08] = 10344738626542428095901441853590
n[09] = 4845764606519362846077015943335
n[10] = -460005211828225354299233114600
n[11] = -476204889207908603032733650067
```

h_{31}

```
d = 1366860518953188827148918565372465257414896071591421359075903127985169665347544101683200
n[00] = -235982689539709975972991465445007491550745230253188468677834392359208605474052005642700
n[01] = -164552547714143411506728270696638716152862363769146482162759461935357887521215447929100
n[02] = -115026484076358557141398516482095289974940960111404727971975811276212185765203277500870
n[03] = -45406665759829249727064809883614224725355289654813822575992874948783748147570751747870
n[04] = 8403057552118811608509887009003120138005627004253164762839887460970105502899456977770
n[05] = 4455395253145158237681350227826603320253844388103827322631047603106423215184838858738
n[06] = -11255937775662311162696769785031672069306921498997614642240548750755974013411441501550
n[07] = -12345442507332585503562864811002971548844180255758377921632756344033911115118078036490
n[08] = -5496255429983792038771807163889590782703880258150495883721863869035353137891049226805
n[09] = -22502308497553752662064734161914421450988809732737277833368503669750173127993161453495
n[10] = 3577349163265773503047826141361266901375947402153818478014443139260590576132012224691
n[11] = 1640917197794259429716225825129027098875393705512955962813041207391469670287038035345
```

h_{32}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = 1794502157397930348560270059462905259167363575646733728269157363848421249917671093424100
n[01] = 878274669574103108138292773339904194354228569093379935437513329506040033604666060335300
n[02] = 67758360158810380608189741819915282944704973832181630187778609191002716191403700689170
n[03] = 308430096879493672906520073288222719414629451851607019222258308315662073245592347677810
n[04] = -53669356837245444612337238145361379813516279667478693071352564345219860037440169519450
n[05] = -23939804551495947596275762026455377683010465970973287671272821335332595051985148584746
n[06] = 5204381858613811638587178793250667426127633091891960563167638706502701285081183180493050
n[07] = 22047754475099947715938905056383520767065690987165467607242761227579431252246226945550
n[08] = 163050710111655736458548717571348623776857673754312661039562482295957622625704603263095
n[09] = 76547138296470857361151708307885097600174839128367506668734900598795189994832738930565
n[10] = -13557456620556791829829995589376826276426127087365390848104140925261681238681052911483
n[11] = -5815567309518269173553618836606318403457991987577815219179286358755136053908807747553
```

h_{33}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = -1447377610144983279640008229442428054362820323207644471453350480203402040899149548980100
n[01] = -575024734890108544356358486022332511202100304452899505077732981570744804920494175492500
n[02] = -519641865078121620134714589149652312809651522820244366516728041868174198030026946417150
n[03] = -223264869216020496674486753907947282237242362727404396873085166147558564371926667499910
n[04] = 39690453600332698295012289042112000869113329017802366161371682771505461448119228846790
n[05] = 17948584686581877698784629136426296001606948822989979332158737113404139232128405777118
n[06] = -373202138208572373539841640065380844238635427343580104726637268856489853377170794996250
n[07] = -167954770763673880808091708847801211829832388995646383572655937614585096709299194580550
n[08] = -141291763281491256052898886039428384684103996199228223041023176251010296114054598596785
n[09] = -58487724999746415037783520816653290077341066323238496504864086170144820509270657977415
n[10] = 10291124334692733838864487167160018830106700043210654356363273073375077641180355674709
n[11] = 4912479391233358337558235999598814149231761334348735825528361757076805411756993532371
```

h_{34}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = 429017948184463648747915493089605304261739526404502553815027978239312417152152624988200
n[01] = 112074872823758624130385490516703931741210571441032371466259042888054285721324536238000
n[02] = 96916848721605194888815497967859982397043190060301480878720810317511921741560516470350
n[03] = 45828014219318055276853244258462463344633272396053234675070915799583392724055816966230
n[04] = -8617217234771045029553866320986158135310528898638224737572343755073113544880202424390
n[05] = -3343552333822704945842345124302513598112512005540401152227251320328060366265769716106
n[06] = 78561652609274082865156362396250951349049448310789096004723917447875046871734982507850
n[07] = 71194196415484124316353828755206935253389870234980997911249896947064691461174566276850
n[08] = 24574222430016640585314834379275038112757184410850742291867220528207536737168754327855
n[09] = 9838558151404444956596523748613599143393804053474664906298423940743007850383553488385
n[10] = -1903209629753064159027818678955039833816394071265687479780639074275510863135475491672
n[11] = -1012814835900131950717158115635257333236125904497332211607419157680241984626926492326
```

h_{35}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = 724890234288883807133692739554181389269667712903527002964305509138092747765250640043000
n[01] = 244047318040953808384618009886615469343407582143500678251608196392283700788171656780960
n[02] = 200583767506299883398799846615839618722912096662345681850356525680047866040867694686370
n[03] = 95072474317732880408873060015477036645940382387376181430971147912620913985476642401730
n[04] = -16547648260093571352064409821528871420122608189302225049240635263600825946450191092942
n[05] = -8665768954945182165273746202102702103834182475211343118142113409291904225223532915282
n[06] = 125097238907996793280228078221895618977339886699390549658898199910084249082893799887550
n[07] = 86899181507597818581114096054078208079345776935909812728525823282898418409948497684310
n[08] = 56847921916204202382738032957377262536116924684885019561903539676492216748841247695345
n[09] = 24547276233504082055846686191137024757811414077289563147946293406387265474531528641755
n[10] = -4337833911022686984336314214527962189832998337927252561590333744868926352043315516740
n[11] = -2128613158185747975129993756707705020446598436320439271687906291812329854772751114454
```

h_{36}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = 243724321419008311235385953077050887168464483820632996153070523023829526776764263685100
n[01] = 19914603719910169529457652233348032267329654498498501534842764254632528988952757336300
n[02] = 99552237782962777945734446575668234192316147426865165066142025736453823983509995947400
n[03] = 41834960618374495916923448606135598267214692703656650871110670698147980108871831462700
n[04] = -7521987311761556960596494538087691612338364731092313670582138591537998935526070960760
n[05] = -3738239736074585498384047768421480280712283333802187493597341120949346362748840653136
n[06] = 171819277401554044884462553306513336223771118192081862103303679190458550351539572469550
n[07] = 28382680992492032227575868910420724874864895485832845639664530387481429244630727082770
n[08] = 22718160367973672521699543570581690080529234476881013406836945389499578569514041890505
n[09] = 11408442049685906353588991812702818440997204486528343055042288115232218634879338057105
n[10] = -2472727747014319016917806964329164875742922562499039620066170921204923787024848410216
n[11] = -783120868471591524536529034452981675286819806984698515943462603720103226019315517636
```

h_{37}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = -659188627283393470512733592499641906893164294771735430705696036614546822723144701270300
n[01] = -259816354186404115859253078296752440755848093868815214321588302892663690875074538393260
n[02] = -191920939487370320371489209019022172362102112967996585430210831768122114773561538100820
n[03] = -91802805894742434779527963429813807393066687458554366569460166694873153403719189608840
n[04] = 15853136588607838984393618921762867076111076420295619602206316508024707173268052486152
n[05] = 6623424020982660277730332140792015589887488810705752924452205083895036950582903034424
n[06] = -145950948389173562104132836109465659836888028605918061856359150113004781191759697996150
n[07] = -81151669743461939948455863961458246302868479179168668437820775234253671339549240852210
n[08] = -59630391970241955172230381490731748445658877169359429225009077650236727470371426246465
n[09] = -2518077749699279537523106581877820357882199977503636714867896861643907090539943585405
n[10] = 4549894098731050180205023458057994500967369588491081787207102240086730571663703467830
n[11] = 2179125837240238013749139260713042491567159230439507073333272015833025166955934105010
```

h_{38}

```
d = 1366860518953188827148918565372465257414896071591421359075903127985169665347544101683200
n[00] = -405310113917233023155099066759884494882095383892066380374158646922700505342214029362300
n[01] = -224688796881114482374122011807054571252704837942491714491694377674215163995276660064020
n[02] = -163473979699209184058490595956602050811094086877074239024149922896029765543020219036920
n[03] = -74764761002985074286621610159799346816226295310098389653631939482447867722199748580760
n[04] = 12128329895439324699579033078429984149039491466884409637883426195825322053199761613394
n[05] = 5763381280089287889010554491825525247290518620235318204143006907127508535690335320842
n[06] = -132768192577360077788703600768295986838194075814118965182734388032934676801232248490550
n[07] = -51158094977006353472143650987227517450020073000833159365822500429898108572652125710990
n[08] = -41368961135045355457012303858045655715565491715515084102446608060521343904598102438915
n[09] = -18960328801446196058826174477909693897513443487320703809362756057796218313686981726905
n[10] = 357308034884702849245324517958032122791617496990149580981899357867351014753149494456
n[11] = 1482827590440617989559770856700765005747159235907346619455537039959534126503513774916
```

h_{39}

```
d = 683430259476594413574459282686232628707448035795710679537951563992584832673772050841600
n[00] = 392663120801943552400896901991220719977390108731924233052889970467736290555117781958000
n[01] = 202045504877653826245704078244601253809701559157461368006861870684162863481805673870960
n[02] = 173185154483668746276181970756186079947428477596572793950067573631171625861338026066750
n[03] = 76948919657316977313367689998831652795894243016658504639882861733206992117144340039370
n[04] = -13605535734670015213895172119244421558827243279858456462998045889027072666316395979092
n[05] = -6067854269979903276078973862181132061117673023085767664554193459298329711890310045796
n[06] = 152003786003313447808793420385613513315762402568179271373568283548903999748792021314300
n[07] = 41080346192109271184710294659061440275348863412584052603094052612152746487579243195300
n[08] = 53742953141309790077928198076107108137353139255578571396508577600613548805649853610700
n[09] = 21857343706088014070490496918952873721913879266235851169180725805617860633164647504660
n[10] = -3919508062073923341517111566683315382619502949671728208536392467092888700431970178443
n[11] = -1703758227107723189941572893055163742852063087979235472354013635557531142341947506755
```

h_{40}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = 784060516836836547859292687798793075468508253100465294911311626203005555432155004159200
n[01] = 355545506877693945495252699684782919421066727691160366523980181823526773144013902221280
n[02] = 357351689009563224248683561165093766780761173691304773517411465560615128255193759978980
n[03] = 150712033782116927109033582777496404684627330145409705400195288941653407985661335039800
n[04] = -2658259599417388810255691599362378439903355443266375965783351962272873318858105984126
n[05] = -12428210033891608060524916593413705862420745920719537998064680227347369798011158765474
n[06] = 241402359082445377828009554152812752956942936649267976308923391849908047743738190302050
n[07] = 65478595920401337424553596239559713614334162588179660803874062777685828564350004364370
n[08] = 79027938473861341699288276193031769084042060386287397049950498289625918876291321629740
n[09] = 35485842819805246584472932764701355764720867082087292555815880038861723992278052938660
n[10] = -6192617036436452593824396039518062498389658866183469284605960400223001441979381981387
n[11] = -2526056175273302533412734508867776741195089466024855315746317446600406443753819943077
```

h_{41}

```
d = 2733721037906377654297837130744930514829792143182842718151806255970339330695088203366400
n[00] = -867460985172622450457702135135638130324762207007270910125532536011042184895666522553800
n[01] = -292620287327156394442613336186684232894186584570672389250923818638318772711787021338160
n[02] = -303929798663737575531101176676386565636200261939825104139285049216250647638106021651220
n[03] = -133274626142982485199378168408754676601829588309386137990449879558470977128204362265620
n[04] = 2410213598308761925572500785350733243886884223381859115728105358845357634688026522862
n[05] = 10576458973711931928171298503172849961093322543667205982816712464177049901723195760198
n[06] = -190431146214427481150390955543135682430150888795121278226181830322708028261117512487100
n[07] = -107276386321317995966276983241429200682171285896767563735490491401800413260442434398740
n[08] = -84491179872614936301522905856482283926798791195047146721689193688286702902462206848175
n[09] = -35308415884570753408879761185014498391484671739208330847129926193804429562312945768975
n[10] = 6003043915001389686424358246179572909778192849235172961508178166250153611618277055418
n[11] = 2874205971376045673249873058714534637886832279103525517937531893559203124662591493880
```

B Rational root boxes and positivity enclosures

Put $N = 2^{256}$. The embedding boxes used for the exact interval checks are

$$\mu \in \left[\frac{M}{10N}, \frac{M+1}{10N} \right], \quad s \in \left[\frac{S}{N}, \frac{S+1}{N} \right], \quad u \in \left[\frac{U}{N}, \frac{U+2}{N} \right], \quad (\text{B.1})$$

where the integer constants are

M = 3491954164956306431453644609980892434381744577868569134761382324120276943749668
S = 258918982791360791136623237694189224638272597174970463806124557733925595847978
U = 440499284018317947084543798940668032418511528266755852350232571514445862522982

The sign tests described in Section 4.3 verify these boxes directly. The enclosures in Table 3 are deliberately rounded outward to simple rational endpoints: each listed integer pair (l, r) means $l/10^{15} < \Delta_{kj} < r/10^{15}$. The strict inequalities follow from the stronger exact rational interval endpoints before outward rounding.

Table 3: Positive leading-minor enclosures, with common denominator 10^{15} .

k	j	l	r
1	1	1913474967234	1913474967235
1	2	3147858252	3147858253
6	1	116844895528629	116844895528630
6	2	692437763721	692437763722
6	3	1179239858	1179239859
6	4	1491101	1491102
7	1	1391263237621	1391263237622
7	2	4228271467	4228271468
7	3	6458386	6458387
8	1	4585428861901	4585428861902
8	2	14054444762	14054444763
8	3	6458386	6458387
9	1	5319184481121	5319184481122
9	2	1906867713	1906867714

All these determinants are positive, establishing positive definiteness without first assuming the validity of an LDL division. Their ratios give the 14 positive pivots. The large integer records in the preceding appendix can thus be interpreted as a positive operator certificate, rather than merely a formal identity over a quotient ring.